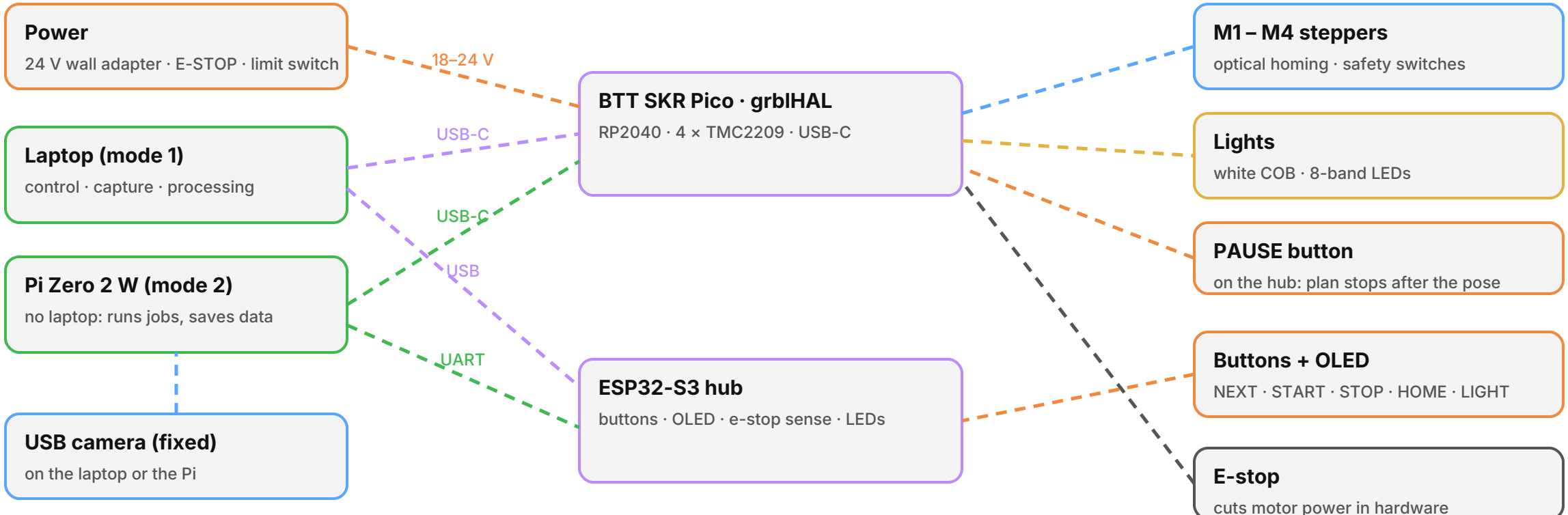
Power
24 V wall adapter · E-STOP · limit switch
Laptop (mode 1)
control · capture · processing
Pi Zero 2 W (mode 2)
no laptop: runs jobs, saves data
USB camera (fixed)
on the laptop or the Pi
18–24 V
USB-C
USB-C
USB
UART
BTT SKR Pico · grblHAL
RP2040 · 4 × TMC2209 · USB-C
ESP32-S3 hub
buttons · OLED · e-stop sense · LEDs
M1 – M4 steppers
optical homing · safety switches
Lights
white COB · 8-band LEDs
PAUSE button
on the hub: plan stops after the pose
Buttons + OLED
NEXT · START · STOP · HOME · LIGHT
E-stop
cuts motor power in hardware